

Para-xylene Production Through Methylation of Toluene

CHE 397– Report #1

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Project Charter

Project: *Para-xylene Production*

Team members: *Mahamat Abdraman, James Fell, Rongwei Lin, William Rawson*

Supervisor: *Adam Brostow*

Goal: *To propose an energy optimized design of a world class para-xylene production plant located in China.*

Key Deliverables:

- A.** *Problem explanation*
 - B.** *Market/Economic analysis*
 - C.** *Solution comparison and selection*
 - D.** *Final design process*
 - E.** *Block flow diagram*
 - F.** *Process flow diagram*
 - G.** *Material Balances*
-

In Scope:

- **Financial justification for new PX plant**
- **Process explanation and description**
- **Comparison of competing processes**
- **Novel energy optimized methylation of toluene**
- **BFD, PFD, and material balances**

Out of Scope:

- **Transportation of product**
- **Safety / EPA regulations**
- **Packaging and distribution**

Design Plan:

William and James – Introduction and appendices, Editing, Research

Mahamat and Rongwei – Process Description, Editing, Research

Introduction

If you were to check the tag on the inside of your shirt, on your favorite bed sheets, or even the curtains draped over your living room windows, there is a high chance these fabrics contain polyester fiber. Even your favorite flick to see at the theatres is made possible by a polyester base coat on the movie film. Polyester fiber has a wide range of application, and in most cases used to improve desirable properties, such as wrinkle resistance in the textile industry [1]. With so many uses, polyester fiber is of high demand, and therefore the components used in its manufacture are as well.

Para-xylene is used as a building block in the manufacture of chemicals sought after in the plastic industry, such as terephthalic acid (TPA), purified terephthalic acid (PTA), and dimethyl-terephthalate (DMT) [1]. Of the three, polyester fiber is made possible by a para-xylene feedstock via the intermediate purified terephthalic acid (PTA) [1]. Generally, para-xylene is consumed as a polymer chain, making it a high value aromatic hydrocarbon. As a member of the BTX group (benzene, toluene, and xylenes), it is extracted from reformat, the product of catalytically reformed naphtha. Of the three possible isomers of dimethylbenzene found in the reformat, para-xylene has methyl groups at the 1,4 position and has the formula $C_6H_4(CH_3)_2$.

A typical para-xylene production plant has the capacity to produce both high purity para-xylene and benzene from a naphtha feedstock, with ortho-xylene as a co-product. At the plant, the naphtha feedstock is hydrated and sent to a separation unit that produces aromatic rich reformat. The reformat includes benzene, toluene, xylenes, and C_9^+ aromatic compounds. With para-xylene being the desired product, it is purified and separated from the rest of the components. With such a big market for para-xylene, the separation and purification steps are designed by chemical engineers to produce high purity para-xylene. However, separation technology can be difficult and is constantly evolving because of the volatility and molecular weight are essentially equal for the xylenes. The separation processes are energy intensive and need to be optimized to produce the most cost-effective process. Furthermore, innovative technologies are able to produce bigger outputs of para-xylene, such as the methylation of toluene described in this report. This additional process can be implemented into a plant to meet future demands in the market, while also creating more revenue from a less valuable toluene by-product.

Economic Trends, Prices, and Demand for Para-xylene

Value Chain

Crude oil contains many types of fuels and hydrocarbon mixtures, like kerosene, natural gas, naphtha and tar for example. Naphtha is the initial feed stock in the production of paraxylene and must be separated from crude oil. When crude oil is distilled, the fraction that boils at 185-350 °F is considered the naphtha fraction. In the petrochemical industry, naphtha is a well-known mixture of heavy and light hydrocarbons. First, it is reformed, or reacted, to convert the hydrocarbons into aromatics and hydrogen before it is separated into its components. For the paraxylene process, it is most beneficial to reform the naphtha to produce the highest number of xylenes possible, which are benzenes with two methyl groups attached. Our desired product is paraxylene; a benzene ring with two methyl groups, one at position 1 and one at position 4.

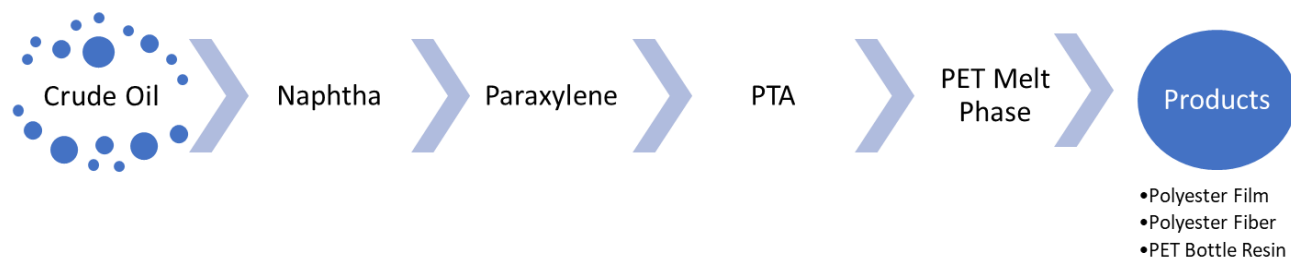


Figure 1: Value chain for the production and end use of paraxylene [1].

Paraxylene (PX) is the most desirable isomer of the xylene family. Around 98% of the paraxylene produced is turned into purified terephthalic acid (PTA) or dimethyl terephthalate (DMT) [1]. PTA and DMT are intermediates that can be turned into polyester fibers, films, or made into polyethylene terephthalate (PET) resin beads (See Figure 2). Other uses for PX include using it as a solvent, turning it into di-paraxylene, and using it in the manufacturing of herbicides. Paraxylene is an intermediate refined petrochemical, making it susceptible to pricing pressure from changes in the price of crude oil and the demand for polyester products (See Figure 1).

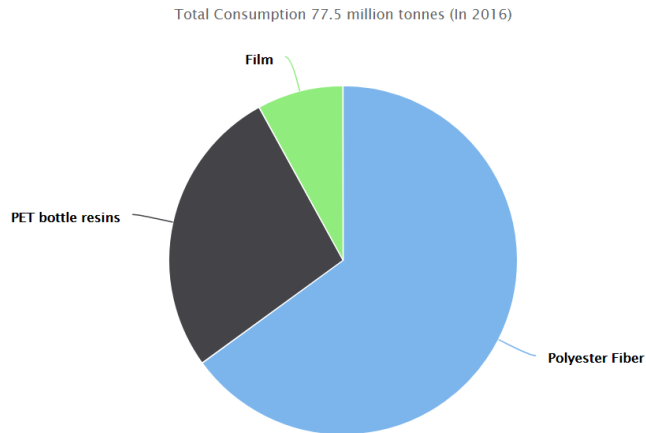


Figure 2: Demand of PTA per application. Application Demand in 2016: Polyester Fiber (65%), PET Bottle Resins (27%), Film (8%). Figure adapted from PlasticInsight.com [2].

Market Drivers

The market for PX has experienced strong growth. This is due to higher consumption rates of PTA, as the polyester manufacturing industry has been increasing output throughout the years. This increase in demand is due to the growth of the polyester fiber industry in Asia, which makes up two-thirds of the global demand (See Figure 3) [1]. In America and Europe, the production of polyester fibers has been declining as the textile industry has been shifting into Asia [1]. Though, in America and Europe, the consumption of PET has been quickly growing as more companies are preferring to switch from glass bottles to PET plastic bottles.

To support the rapidly growing textile industry in Asia, around 8.5 million metric tons per year of additional PX production capacity was commissioned in Northeast Asia in 2014 [3]. The commissioning of the new capacity caused an oversupply of PX and put downward pressure on the profitability of producing PX. This ultimately led to an oversupply of 10 million tons of PX and forced companies to decrease operating rates to less than 80% [4,5]. Since then, steady growth in the consumption of PX and limited production of PX has greatly depleted the excess PX capacity (See Figure 4). The global demand for PX has been growing at a quick pace and is expected to continue as the utilization of more PET bottle resins and polyester fibers find more uses in the global industry. Current projections from OrbiChem predict the average global consumption of PX to grow at 7% per year for the next 10 years [2]. Growth in PX consumption will be primarily driven by China's increased imports of PX, which is expected to be greater than 10 million tons per year [5].

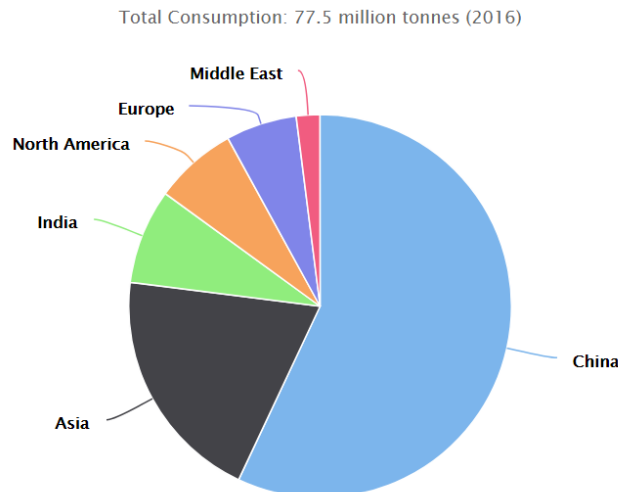


Figure 3: Demand of PTA per region. Market Demand in 2016: China (57%), Asia (20%), India (8%), North America (7%), Europe (6%), Middle East (2%). *Figure adapted from PlasticInsight.com [2].*

Feed Stock and Product Pricing

The production of paraxylene relies on a both naphtha and a methanol feed stock for the conversion of toluene to paraxylene. The valuable products and byproducts in the production of paraxylene include paraxylene, orthoxyelene, benzene, and hydrogen. The most important metric to focus on is the price spread between the naphtha feedstock, methanol price, and the selling price of the main product, paraxylene. Current prices show that naphtha sells for \$500 USD per metric ton, methanol from Methanex corporation sells for \$335 USD per metric ton, and paraxylene sells for \$1050 USD per metric ton [5,6]. The cost spread between the naphtha feedstock/methanol feed stock and PX is upwards of \$215 USD per metric ton, which makes the production of PX still quite profitable, even with the added expense of methanol. Current 2020 market prices for the byproducts of our process show that Benzene sells for \$645.33 USD per metric ton, orthoxyelene sells for \$947.99 USD per metric ton, and hydrogen sells for \$700 USD per metric ton [5]. Prices have been slowly falling over the past few years as an oversupply of PX still exists and oil prices have remained low. Factoring in the added profits from the sale of the byproducts, this process can be very profitable, even in an over supplied economy. The oversupply in PX is quickly diminishing though, so plant utilization will only continue to increase from here on. More PX plants will have to be built to meet future supply and current plants may have to increase the conversion of naphtha to PX to increase profitability/production capacity.

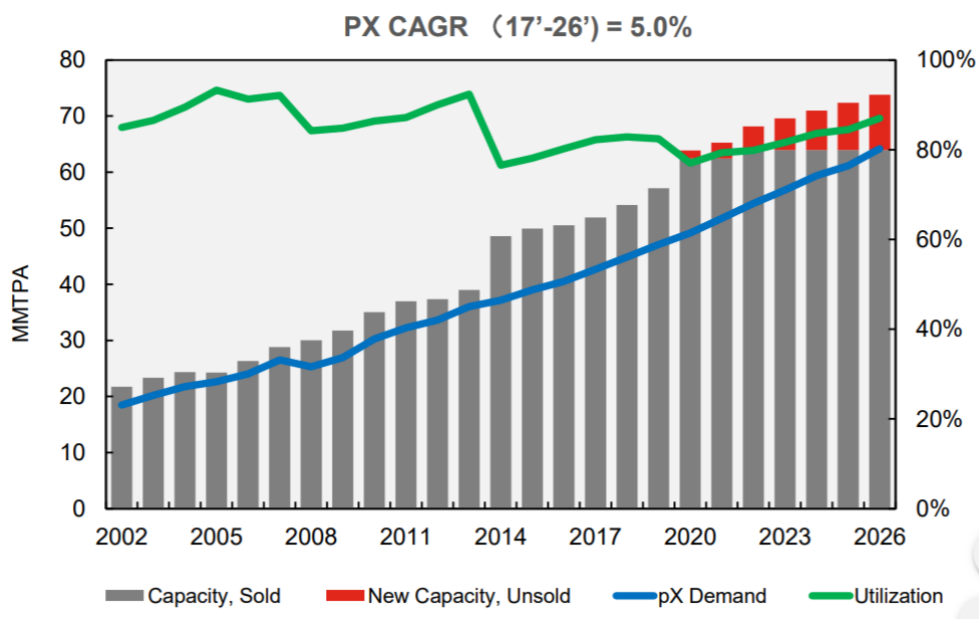


Figure 4: Global demand, capacity, and utilization of PX. *Figure adapted from Advances in Aromatics Technology by UOP Honeywell [5].*

Regulatory Environment

The production of paraxylene produces benzene as a valuable byproduct of the process. Benzene is an extremely carcinogenic chemical and in 2002, China imposed stricter regulations for permissible benzene concentrations in the air. They reduced the permissible level of benzene in the air from 10 ppm to 1.8 ppm [7]. Although these regulations are much more generous than in the United States, the stricter regulation will require higher costs associated with more safety equipment, safety inspections, and higher tolerances for equipment. To meet these regulations, higher capital costs can be accrued in the manufacture, storage, handling, and shipping of benzene. Other regulations imposed by China include capping the capacity on the amount of paraxylene produced in the region to maintain stability and profitability in the manufacturing of paraxylene. The paraxylene capacity cap fluctuates based on the current supply and demand. Currently most paraxylene plants can only operate at 60-80% capacity, depending on the output capacity of the plant [5]. We can expect that if we build a world class paraxylene production plant, we can expect to operate at 60% for the first few years and then slowly increase capacity as demand increases in China and oversupply begins to decrease.

Competitors

As of 2018, the world's largest producers of PX were Hengli Petrochemical, Reliance Industries, SINOPEC, and ExxonMobil [8]. The world's largest producer of PX is Hengli Petrochemical, which has a total output of 4.5 MMTPA. Reliance Industries is the second largest producer and currently has a facility that produces 4.3 MMTPA of PX. ExxonMobil currently has a facility that produces 3.5 MMTPA of PX. SINOPEC has a facility that produces 1 MMTPA of PX. It is predicted that by 2023, the Middle East will become the third largest producer of PX in the world. Their heavy investments into local feedstock integration, large-scale production facilities, and access to low cost energy will make the Middle East very cost competitive on the global market [8]. The key to constructing a profitable PX process with the ability to compete with the world's leading producers of PX will be to make the production of PX extremely cost competitive. This will include minimizing energy costs and maximizing the conversion of naphtha to paraxylene.

Goals and Objectives

After obtaining a more detailed look into the para-xylene industry from the market analysis, clear goals were drawn out to achieve a profitable world class para-xylene production plant. The purpose of this report is to provide an innovative and novel design for the production of para-xylene. The design will implement existing technological models, such as the methylation of toluene, while also incorporating new energy optimization ideas that will still allow for the production of para-xylene of optimal purity. Similar processes have already been modeled and commercialized by ExxonMobil, but by performing a heat integration of the highly exothermic reaction process, energy consumption will be reduced. Less energy consumption, along with lower capital costs from the avoidance of more complex separation equipment, will make the model more economically viable and more environmentally friendly. In addition to saving costs on lower utilities, less CO₂ will be emitted from the plant, helping the process fight back against pollution. Furthermore, this plant is designed to keep up with market demand and allow for the fluctuations of current product prices. For example, toluene and methanol have lower costs than what is made through the sale of para-xylene, making the methylation of toluene a good alternative to the more fundamental process of turning toluene into benzene.

Process Description

The conventional process of producing paraxylene involves the reforming of naphtha from petroleum distillate. Generally, the distillate is a narrow cut of naphtha, meaning most of the lighter hydrocarbons have been removed. The narrow cut of naphtha contains C_7^+ hydrocarbons which convert naphthenes (a group of cyclic aliphatic hydrocarbons) to aromatic hydrocarbons through the reformation process [9]. Such processes are designed and operated to maximize the yield of the desired aromatic product, paraxylene, from the naphtha feed [9].

To begin the conventional process, naphtha in the reforming unit is contacted with a strong hydrogenation metal catalyst to produce reformate, hydrogen gas, and fuel gas. The reformate generally contains BTX, C_9^+ aromatic hydrocarbons, and ethyl benzene [9]. To obtain a purer liquid reformate, the hydrogen and fuel gas need to be removed, and are done so in the reforming and stabilizer units. The liquid reformate contains paraffins, ethyl benzene, BTX, and C_9^+ aromatic hydrocarbons, and is sent to a splitter to remove the C_9^+ aromatics. The product of the splitter, containing paraffins, ethyl benzene, and BTX, is then sent to an extraction unit in which a selective solvent is used to extract the undesirable aromatics and unconverted paraffins [9]. The product is a mixture of aromatic compounds composed of BTX, ethyl benzene, and small amounts C_9^+ aromatics. Next, the effluent mixture is sent to a fractionation unit in which benzene and toluene are removed as the distillate. The bottoms of the fractionation column contain xylenes and C_9^+ aromatics, which are then sent to a xylene column to separate out the desired para-xylene product [9]. The distillate is then fed to another column to separate benzene from toluene to obtain a benzene product. In bottoms of the benzene column, the toluene is sent to a transalkylation / toluene disproportionation unit [9]. Benzene, xylenes and C_9^+ aromatics are further produced in the toluene disproportionation unit in the presence of hydrogen through the contact of an acid zeolite catalyst [9]. The effluent, containing mostly xylenes and C_9^+ aromatic hydrocarbons, are sent to a xylene rerun column where the heavy aromatics are removed, and the para-xylene rich aromatics are sent to a para-xylene recovery unit. In the para-xylene recovery unit, some para-xylene is recovered, while the para-xylene depleted stream is sent to an isomerization unit [9]. Fractional crystallization or selective adsorption (PAREX) is usually used to operate the para-xylene unit for separation. The para-xylene depleted stream is then fed to a xylene isomerization unit where the xylene isomers are isomerized back to their equilibrium concentrations. Furthermore, some of the aromatics are converted to ethane,

benzene and diethylbenzene by isomerization. This is done to obtain a larger production of xylenes [9]. The equilibrium aromatic mixture is then recycled back to the para-xylene recovery unit after removal of any undesired C_9^+ aromatic by-products to further recover para-xylene [9].

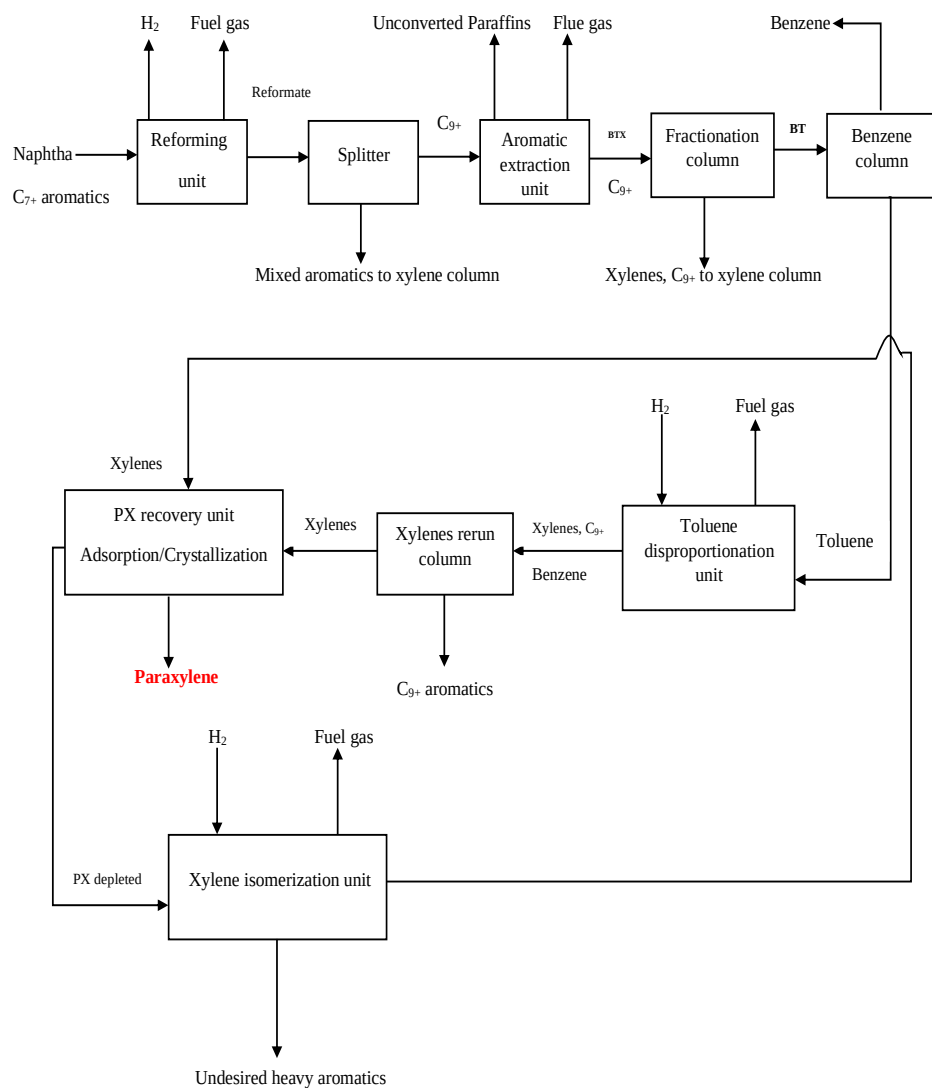


Figure 5: Block flow diagram for conventional paraxylene process from raffinate naphtha [10].

Possible solutions

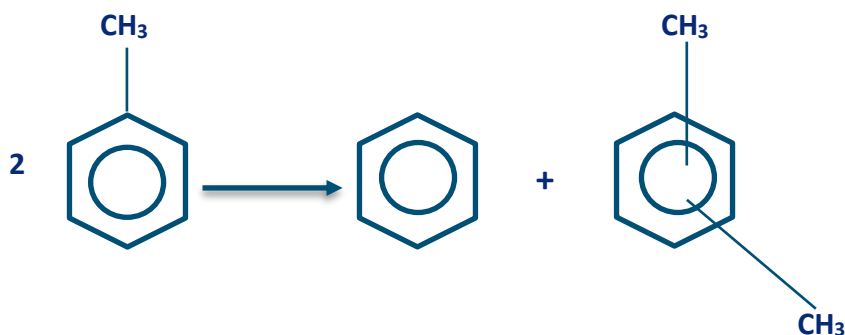
The production of para-xylene from naphtha feedstock requires a rigorous implementation of technological models to achieve a high-quality para-xylene product. A production method that produces high purities, high recoveries, and a flexible separation of para-xylene from its isomers is the most desirable operation. To achieve such a target, some of the competing industry methods are UOP Honeywell's process and ExxonMobil's process. Both processes operate in similar ways to the conventional process defined in the description above [see process description page 11]. However, differences show up in how toluene is converted into more valuable products and how para-xylene is separated out from its other isomers. UOP uses a toluene disproportionation process as a production method to convert toluene into benzene and xylenes. They also use a proprietary unit called PAREX as a means of separating the xylenes by a selective adsorption purification method. In contrast, ExxonMobil uses a toluene methylation production method involving alkylation of toluene with methanol in a bed reactor. The purification process is achieved through a reactive distillation unit. However, both UOP's and ExxonMobil's methods use a similar zeolite catalyst to convert toluene to para-xylene, making both processes more profitable [10].

1. UOP LLC

Production method:

In the case of toluene disproportionation, benzene and xylene are formed by a reaction of toluene over an acid zeolite catalyst [10]. The implementation of such technology requires three main processing areas. There is a reactor section where mixed aromatics and benzene are produced in the presence of hydrogen and a catalyst, a product distillation section where undesired aromatics are removed, and a para-xylene recovery section where xylenes are recovered and separated [10]. Crystallization or adsorption separation is typically used with this model to achieve at least 90% selectivity of paraxylene in the xylene mixture [10]. One of the disadvantages of these processes is that extensive separation is needed for high para-xylene purity due to low selectivity from the disproportionation unit. Moreover, such a process is only economically feasible when the demand of benzene is as high as that of xylene in the market [10].

The main toluene disproportionation reaction is the following:



Purification method:

UOP has developed and commercialized an adsorption purification method involving a proprietary solid selective adsorbent called ADS-37. This adsorbent is used in the PAREX process unit and is capable of achieving a purity of 99.9 wt% para-xylene at greater than 97% recovery [11]. Detailed information of the PAREX operation is not publicly available due to its sensitivity. However, separation of para-xylene from its isomers can be achieved by selective adsorption on a molecular sieve. This is possible due to the different positions of the methyl groups found in xylenes. The different positions of the methyl groups give different kinetic diameters: $6.7 \text{ \AA}$ for p-X, $7.3 \text{ \AA}$ for o-X and $7.4 \text{ \AA}$ for m-X [10]. In other words, such differences result in a relatively high diffusion coefficient of para-xylene compared to those of other isomers[10]. Furthermore, separation is accomplished by exploiting the differences in affinity of the adsorbent for paraxylene relative to the other C_8 isomers on zeolites catalysts [10]. As a result, a recovery of 99.9 % pure para-xylene is possible in the extract while a mixture of its isomers is withdrawn in the raffinate [10].

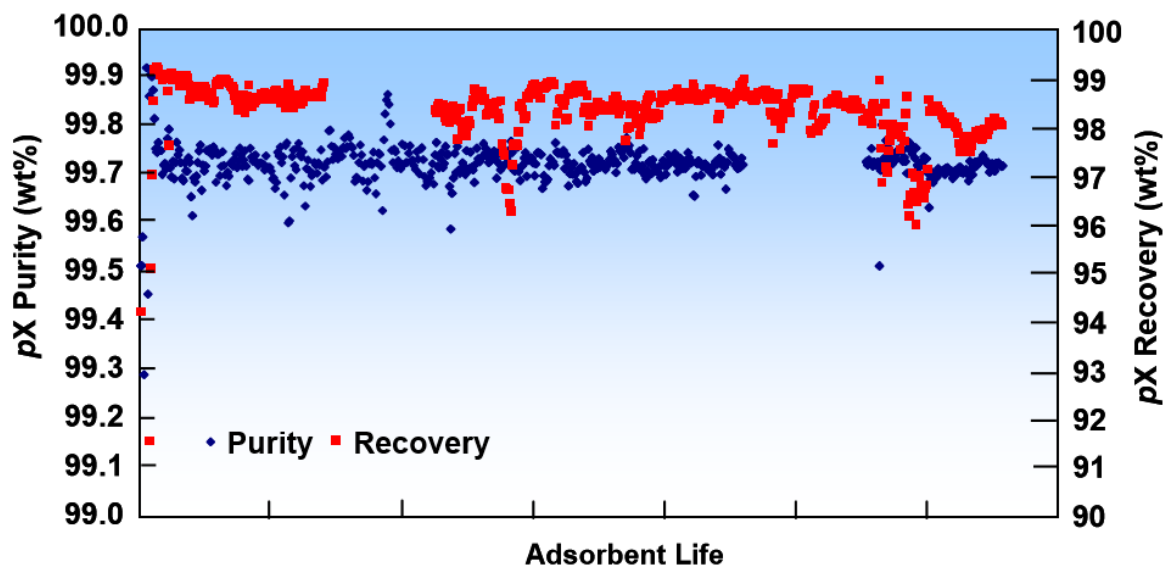


Figure 6: Commercial PAREX unit performance, in terms of purity and recover, with ADS-37 adsorbent for over 2+ years [11].

The PAREX unit facilitates the revamping of equipment to maximize production capacity. However, this makes the overall project long, with the cost of revamping being relatively high [11]. Hence, the UOP suggested process requires extensive implementation of complex units that would lead to high energy cost even though it results in highly pure para-xylene with high recovery.

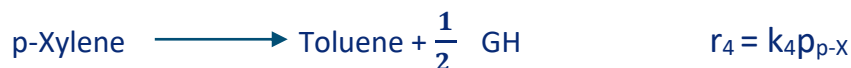
Toluene methylation:

Production method:

Methylation of toluene is a more economically feasible process to produce para-xylene if the cost of methanol remains relatively low [10]. The main reaction used in this process consists of toluene and methanol over a zeolite catalyst ZSM-5. This reaction produces water and xylenes, with a thermodynamic equilibrium composition of xylene isomers being 23.55% p-xylene, 52.42% m-xylene and 24.03% o-xylene at around 400 °C [10]. However, different modifications of zeolite catalysts result in higher para-xylene selectivity which makes methylation a suitable alternative. Para-xylene selectivity of 90% or higher is possible with boron or magnesium doped zeolite catalyst [10]. In fact, boron modified catalyst exhibits a para-xylene selectivity higher than 99.9%, but the loss of boron

from the catalyst during reaction makes it unstable [10]. Alternatively, a magnesium modified zeolite catalyst gives a para-xylene selectivity of 92.7%, while minimizing the contact time on external acid sites. This selectivity is achieved by the decrease in residence time, which prevents paraxylene isomerization [10]. It is possible to obtain an even lower residence time by adding N₂, or H₂O to the feed of the reactor, thereby increasing selectivity further.

Optimization of reactor operating conditions, such as pressure and temperature, can also increase para-xylene selectivity. The methanol is completely converted at a temperature ranging from 400 °C to 560 °C [10]. There are five reactions for the process of toluene methylation over Mg-modified ZSM-5 catalyst. This includes a main reaction, with four side reactions: toluene methylation, methanol dehydration, toluene disproportionation, paraxylene dealkylation and paraxylene isomerization. The rate expressions are given in terms of partial pressures 'p' and rate constants 'k'.



Where GH indicates gaseous hydrocarbons such as methane, ethane, ethylene, propane and butanes, which is represented as pure ethylene to ensure stoichiometric balance.

Toluene methylation is more favorable because its mild operating conditions, such as a pressure of 3-4 bar and a temperature of 400-500 °C, result in lower utilities [10]. Moreover, the more expensive implementation of separation units, such as the selective adsorption PAREX unit, are not needed for this method. Optimization of the process can be accomplished by analyzing the effects of parameters in such as feed composition, feed ratio, residence time, temperature and pressure. This work can be done in ASPEN plus using proper design specifications, property method and operating conditions, with the end goal of achieving a para-xylene selectivity as high as 99.7%

[10]. All things being considered, the methylation of toluene production method is more advantageous in that it produces a higher yield that facilitates the separation of para-xylene from its isomers.

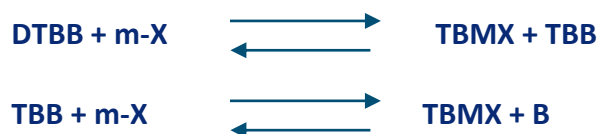
Purification method:

Conventional distillation processes for separating different components requires a significant difference in boiling points or a relative volatility of greater than 1. With this criteria, benzene and toluene can be traditionally separated due to relative volatility of 3.09. Para-xylene can't be separated though from its isomers using a simple distillation unit as their boiling points are very close and their relative volatilities are close to 1. Moreover, the presence of isomer components doesn't facilitate para-xylene separation due to the similar molecular structures of the isomers [10]. As a result, a rigorous distillation unit, such as reactive distillation column, is required to carry out a separation of paraxylene from m-xylene and o-xylene [10]. Alkylation of the xylene mixtures in a distillation unit makes separation of para-xylene from m-xylene possible [10]. Thus, m-xylene can selectively react with di-tertiary butyl-benzene (DTBB) and tertiary butyl-benzene (TBB) to form tertiary butyl m-xylene (TBMX) and benzene (B) [10]. Such reactions lead to a higher boiling point of tertiary butyl m-xylene that in turn becomes easier to separate from para-xylene through fractionation [10]. The normal boiling points of the individual components are shown below for comparison.

Table 1: Normal boiling points of components for the reactive distillation

Component	Normal boiling point, °C
p-xylene	138.3
m-xylene	139.1
o-xylene	144.4
TBB	169.3
TBMX	205
DTBB	230

The reaction in the reactive distillation column is rapid and immediately reaches equilibrium. The reactions are as follows:



Benzene is quickly separated because it's the lightest of all components in the reactive distillation unit [10]. Sensitivity analysis can be used in an ASPEN simulation by implementing proper design specifications and varying parameters to achieve a paraxylene purity of 99.7%. Hence, this separation method is more favorable because it is conventional and avoids extensive energy consumption.

Chosen solution

For our design, we chose to implement the methylation of toluene method to produce para-xylene. In this method, an aromatic stream containing mostly toluene is preheated to a predetermined temperature, along with a fresh feed stream containing methanol. These streams are mixed, heated again, and then supplied to a reactor containing an alkylation catalyst. By controlling the feed stream temperatures, the molar ratio of methanol to toluene, and the temperature in the reactor, the selectivity of the reaction is maximized, along with the para-xylene product yield.

In the overall para-xylene production process, a liquid reformat is obtained from naphtha feed stock in a catalytic reforming unit. The reformat is then stabilized by expelling out the fuel gas in the mixture and sending the remaining aromatics to a splitter. The C_9^+ aromatics are removed in the bottoms of the splitter, while the tops are sent to an aromatics extraction unit. In the extraction unit unconverted aromatics of C_{20} and higher are removed at the top, along with some left over fuel gas impurities. The remaining BTX, ethyl benzene and C_{9+} impurities are sent to a benzene column. In the benzene column, benzene is removed as the distillate product, while the toluene rich stream is sent to a toluene methylation unit for further processing. In the toluene methylation unit, a toluene feed and a fresh methanol feed (which contains water to suppress side reactions) are first preheated and vaporized in a furnace before being sent to a fixed bed reactor for an alkylation reaction in the presence of magnesium doped ZSM-5 zeolite catalyst. In the alkylation reactor, toluene reacts with methanol to produce para-xylene. Side reactions also occur, such as methanol self-decomposition into gaseous hydrocarbons and water, and para-xylene conversion to m-xylene and o-xylene. The

reactor effluent is then cooled down and sent to a flash drum where light compounds, such as gaseous hydrocarbons, are sent to the overhead. The bottom liquid is fed into a decanting vessel for further separation. In the decanting vessel, the upper layer product is water and the lower layer is an aromatic mixture containing benzene, toluene, p-xylene, m-xylene and o-xylene, among other heavier aromatics. The aromatic mixture then goes through a series of separations in which it is first sent to a benzene column, where the benzene distillate is recovered at the overhead. The bottoms, containing toluene and xylenes and aromatics, is then sent to a toluene column. Toluene is recovered as the distillate and recycled back to the reactor mixed feed. The bottom stream containing xylenes and mixed aromatics is then sent to a reactive distillation unit where 99.7 wt % of PX is recovered at the top of the reactive column [10]. In the reactive distillation columns, the xylene isomer m-xylene is reacted with ditertbutylbenzene in the presence of aluminum chloride to form tertbutyl xylene, which can be separated from p-xylene due to high volatility. It is sent to the top of the column while the remaining mixed aromatics is released at the bottom of the column [10].

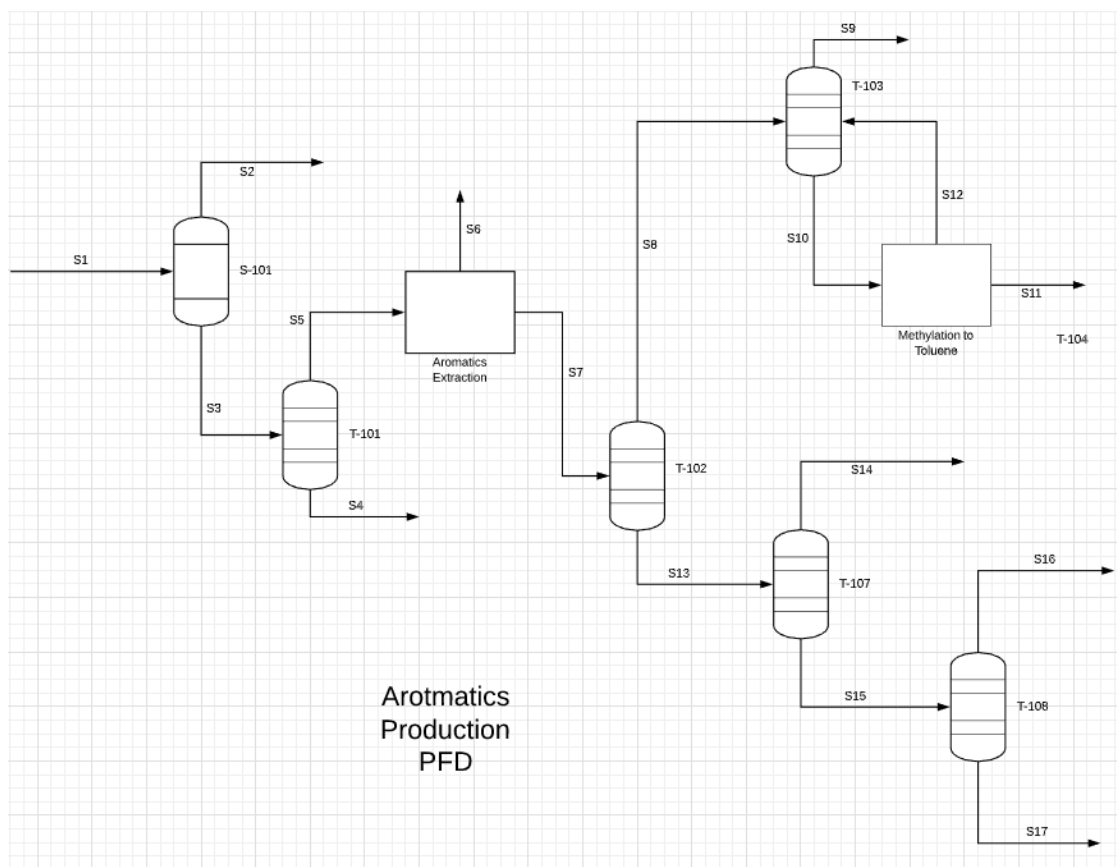
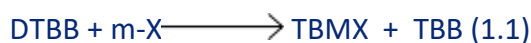


Figure 7: PFD of Aromatics Production through methylation of toluene process

The relative volatility of benzene / toluene and o-xylene / para-xylene were 3.09 and 1.17, respectively. Benzene can be easily separated from the mixture by distillation, and the separation of o-xylene requires a large amount of distillation due to its low relative volatility. It is difficult to separate para-xylene and meta-xylene by ordinary distillation, because the boiling points of the two components are too similar and their relative volatility is only 1.02. Due to the similar molecular structure, they are not affected by the presence of the third component, and extractive distillation does not work. Para-xylene and meta-xylene can be separated by reactive distillation, as reported by Saito et al. [12]. This method involves alkylating a mixture of xylenes. M-xylene reacts preferentially with di-*t*-butylbenzene (DTBB) and *t*-butylbenzene (TBB) to form *t*-butyl-*m*-xylene (TBMX) and benzene (B). The reaction is shown in Equations 1.1 and 1.2 [12]. M-xylene forms tertiary-butyl *m*-xylene with a higher boiling point, which can be easily separated from *p*-xylene by fractional distillation.



The reaction occurs quickly and reaches equilibrium immediately [13,14]. The benzene produced in the alkylation reaction is the lightest and is distilled off, which can disrupt the equilibrium. Equilibrium data are given by Saito et al. [12] And the reactions in equations (1.1) and (1.2) are shown in equations (1.3) and (1.4), respectively.

$$K^R = \frac{x_{\text{TBMX}} x_{\text{TBB}}}{x_{\text{DTBB}} x_{\text{m-X}}} = 0.6 \quad (1.3)$$

$$K^R = \frac{x_{\text{TBMX}} x_{\text{B}}}{x_{\text{TBB}} x_{\text{m-X}}} = 0.16 \quad (1.4)$$

Toluene methylation is the most optimal para-xylene production process for it requires less complex and less extensive separation columns such as crystallization or adsorption (PAREX). The reason is that the methylation reaction exhibits a higher yield that facilitates the implementation of conventional distillation units. It is economically more viable and environmentally friendly than existing processes, considering that it uses less unit operations that contribute to lower utility costs.

Advantages for the Solution

High purity of para-xylene

One of the advantages of this technology is that it can produce xylene products rich in para-xylene through a bed reactor, thus greatly reducing the expectations and material consumption of the separation and alternation process. Alkylation of toluene and methanol is a new process for para-xylene production with greater economic benefits, which can maximize the utilization of toluene. The toluene alkylation method can maximize the production of valuable para-xylene products while reducing feed and energy costs. The high paraxylene selective process is based on commercially proven technology. Unlike other toluene conversion processes, it has no benzene by-products and no co-feed of hydrogen. It is the only method that can adjust the methyl ring ratio according to market conditions and allow unrestricted benzene co-feed to produce para-xylene [15].

Less equipment

In this method, the use of the distillation apparatus can reduce the use of the toluene disproportionation unit and parex unit. These two units are very expensive, so this a more economical apparatus. Furthermore, it has lower feed and energy costs [15]. In the distillation unit, the monitoring equipment of temperature and pressure is very sensitive. Therefore, we can detect and prevent hazards at quicker rates. The process is more stable and has consistent product yields.

Environment friendly

During the production of para-xylene process, no benzene co-product and hydrogen co-feed are produced. As a result, this is an environmentally friendly process. Because the solution has continuous regeneration of para-xylene, it has low sensitivity to poisons.

Appendices

A1.) - Design Basis

We propose building a world class paraxylene production plant in Dalian, China. We expect to produce 4.5 MMTPA of paraxylene with a purity 99.9%. With the implementation of the methylation of toluene to paraxylene, we are targeting a recovery of 99% of paraxylene from the reformed naphtha. We plan to have the construction of the plant done by 2025, with a 25% operating rate the first year, a 50% operating rate the next year, and then decide future operating rates based on the current year paraxylene demand.

Plant Capacity

The consumption rates of paraxylene globally are quickly increasing and are creating an opportunity to capitalize on the need for increased paraxylene production in the future. The largest producers of paraxylene are Hengli Petrochemical and Reliance Industries, which have facilities that produce 4.5 MMTPA and 4.3 MMTPA of PX, respectively [16,17]. To be competitive against the world's largest producers of PX, we need to the ability to match their production rates. Larger production rates also give room to decrease or increase production rates based on the current PX economy. A larger production facility will also create higher revenues that should help offset times of low PX prices. Therefore, we choose to build a world class 4.5 MMTPA PX facility.

Purity

Most PX production facilities target a purity of 99.5-99.7 wt% with a recovery greater than 98% [18]. The market is now demanding purities of PX that are 99.9 wt% and above [19]. This is due to the shift of higher quality PTA and its use in higher quality polyester fibers. For our PX plant, we will target a PX purity of 99.9 wt% and a recovery of greater than 99%. Increasing the recovery of PX from the reformed naphtha will increase the profit margin as more naphtha is being turned into PX instead of by-products.

The System of Units to be Used

The system of units that our plant will be design with will be the metric system. Our plant will be based in China, so for ease of communication with Chinese contractors, the metric system will be used.

Details of Raw Materials that are Available

The raw materials that are needed in our proposed production of PX is naphtha and methanol. Currently, China is importing large amounts of naphtha with very little exports of naphtha [20]. In 2019 Zhejiang Petrochemical started up its naphtha cracker, which required 200,000 metric tons per month of naphtha [21]. This caused naphtha imports to rise in China with the additional consumption by Zhejiang Petrochemical. Also, China is currently the largest importer of methanol, with China accounting for 22% of worlds imports in 2018 [22]. China's large industrial sector consumes more naphtha and methanol then it can produce, leading it to import large amounts of the commodities. The addition of our world class PX production facility will ultimately increase the amount of naphtha and methanol being imported into the country.

Location of Plant

The location of our proposed paraxylene production facility will be built in Dalian, China (See Figure 1). China's demand for paraxylene has been growing quickly over the past few decades that China is the largest manufacturer of polyester fibers in the world. Building our plant near the costumer will reduce transportation costs and support the heavy industrialization that is taking place in China. The town of Dalian, more specifically, is the prime location for our plant. In Dalian, the largest producers of PTA are located in Dalian. The Hengli Petrochemical plant and Yisheng Dahua Petrochemical Company are located in Dalian [16]. This gives our plant very close proximity to potentially very big costumers of PX. Another advantage of Dalian is being a peninsula, shipping routes can easily be implanted form the ports of Dalian to the major ports along China's trade routes. The introduction of a new PX production facility will require imports of methanol and naphtha, shipping of paraxylene and other by-products. Being located near the ports of Dalian will save on transportation costs for both import and exporting goods. Other advantages of choosing Dalian as location of our plant is the

availability of many petrochemical infrastructure resources and low cost of utilities (see following Utilities section) [16].



Figure 8: Location of our proposed paraxylene production facility. *Figures adapted taken from Google Maps ©.*

Utilities

The production of paraxylene requires electricity, fuel gas, and cooling water to power, cool, and heat the process equipment involved. In 2019, the price of electricity for business in China was \$0.097 USD per kWh [23]. The Chinese government has made it a top economic importance to make a unified nation grid system to improve the reliability of electricity in the country [24]. In Dalian, China, the current industrial infrastructure helps mitigate any electrical reliability risks, although a general risk of reliability still exists. In January 2020, the current price of natural gas, or fuel gas, was \$0.38 USD per m³ of natural gas [25,26]. China currently imports large amounts of liquified natural gas (LNG) to meet its growing demand for natural gas. Consumption of natural gas is growing rapidly in China making it hard for suppliers to meet demand as China's energy grid is still in developing [26]. As industrialization in China continues to increase, LNG imports will have to increase in order for our PX plant to have a reliable supply of natural gas. An estimate for the cost of process water is to assume that it takes approximately 2 kWh for a recirculating cooling water system to circulate 1000 gal of water [27]. It is also estimated that it costs \$0.02/1000 gal of water to account for the cost of the water make-up and chemical treatments [27]. This gives us a cost of approximately \$0.214/1000 gal for process water.

A2.) – BFD & Material Balance

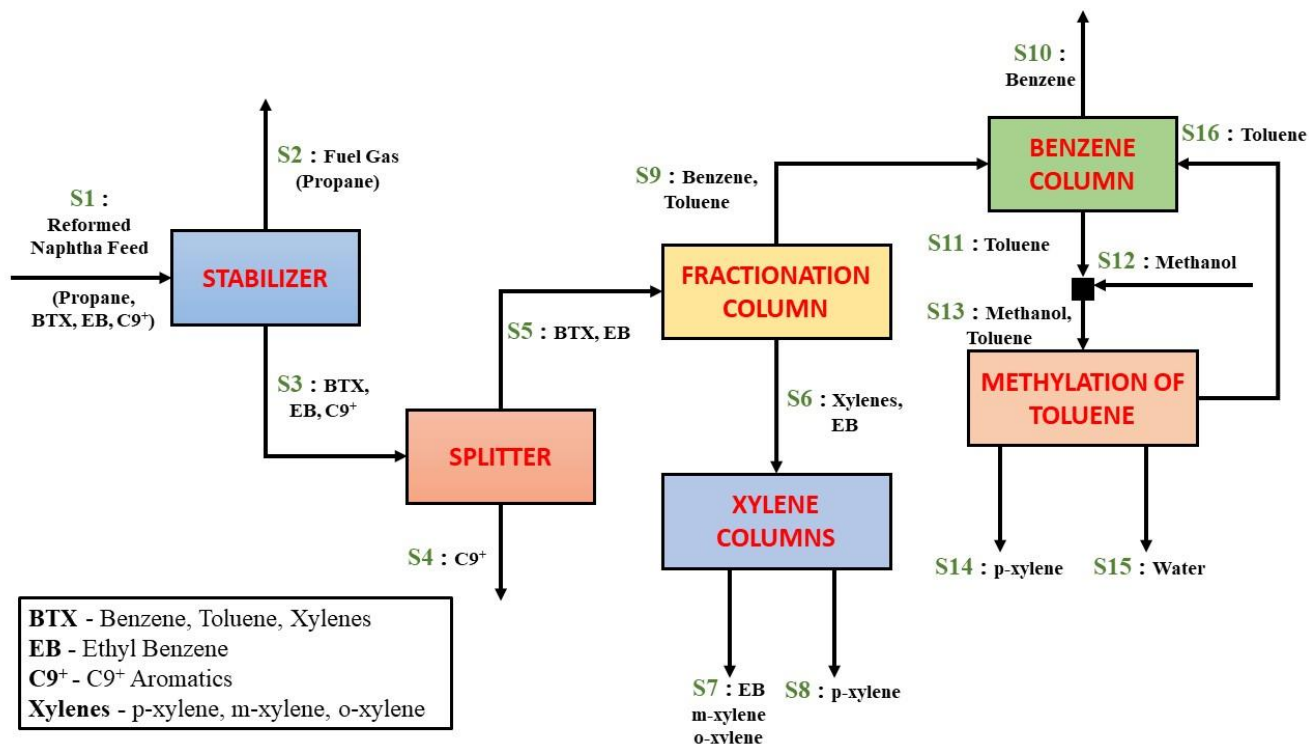


Figure A2-1: BFD of Para-xylene production through the methylation of toluene.

Table A2-1: Material balance of the para-xylene production process through the methylation of toluene.

	S1 - Naphtha Reformate Feed		S2 - Stabilizer (Top)		S3 - Stabilizer (Bot)		S5 - Splitter (Top)		S4 - Splitter (Bot)		S9 - Fractionation Column (Top)		S6 - Fractionation Column (Bot)	
Species	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)
Propane	1.0%	15272	100.0%	15272	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
Benzene	10.1%	154601	0.0%	0	10.2%	154601	13.4%	153055	0.4%	1546	26.4%	151525	0.3%	1531
Toluene	27.8%	424493	0.0%	0	28.1%	424493	36.7%	420248	1.2%	4245	72.6%	416046	0.7%	4202
ETHYLBENZENE	6.6%	100688	0.0%	0	6.7%	100688	8.7%	99681	0.3%	1007	0.2%	997	17.3%	98684
p-XYLENE	6.8%	104167	0.0%	0	6.9%	104167	9.0%	103126	0.3%	1042	0.2%	1031	17.9%	102094
m-XYLENE	15.2%	232427	0.0%	0	15.4%	232427	20.1%	230102	0.6%	2324	0.4%	2301	39.9%	227801
o-XYLENE	8.9%	135291	0.0%	0	8.9%	135291	11.7%	133938	0.4%	1353	0.2%	1339	23.2%	132599
C9+ (TMB)	23.6%	360299	0.0%	0	23.8%	360299	0.3%	3603	96.9%	356696	0.0%	36	0.6%	3567
SUM:	100.0%	1527238	0.0%	15272	100.0%	1511966	100.0%	1143754	100.0%	368212	100.0%	573275	100.0%	570479

	S10 - Benzene Column (Top)		S11 - Benzene Column (Bot)		S12/S13 - Methylation of Toluene (In)		S15/S16 - Methylation of Toluene (Out)		S14 - Methylation of Toluene (PX)		S7 - Xylene Separation (EB,MX,OX)		S8 - Xylene Separation (PX)	
Species	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)
Water	0.0%	0	0.0%	0	0.0%	0	78.2%	78153	0.2%	789	0.0%	0	0.0%	0
Methanol	0.0%	0	0.0%	0	25.5%	143226	2.8%	2836	0.0%	29	0.0%	0	0.0%	0
Benzene	97.3%	150009	0.4%	1515	0.3%	1515	1.5%	1500	0.0%	15	0.3%	1515	0.0%	15
Toluene	2.7%	4160	98.3%	411885	73.3%	411885	8.2%	8155	0.0%	82	0.9%	4160	0.0%	42
ETHYLBENZENE	0.0%	10	0.2%	987	0.2%	987	1.0%	977	0.0%	10	21.0%	97697	0.9%	987
p-XYLENE	0.0%	10	0.2%	1021	0.2%	1021	4.7%	4661	99.8%	461427	0.2%	1021	95.6%	101073
m-XYLENE	0.0%	23	0.5%	2278	0.4%	2278	2.3%	2255	0.0%	23	48.5%	225523	2.2%	2278
o-XYLENE	0.0%	13	0.3%	1326	0.2%	1326	1.3%	1313	0.0%	13	28.2%	131273	1.3%	1326
C9+ (TMB)	0.0%	0	0.0%	36	0.0%	36	0.0%	35	0.0%	0	0.8%	3531	0.0%	36
SUM:	100.0%	154227	100.0%	419048	100.0%	562274	100.0%	99885	100.0%	462388	100.0%	464721	100.0%	105757

A3.) – PFD

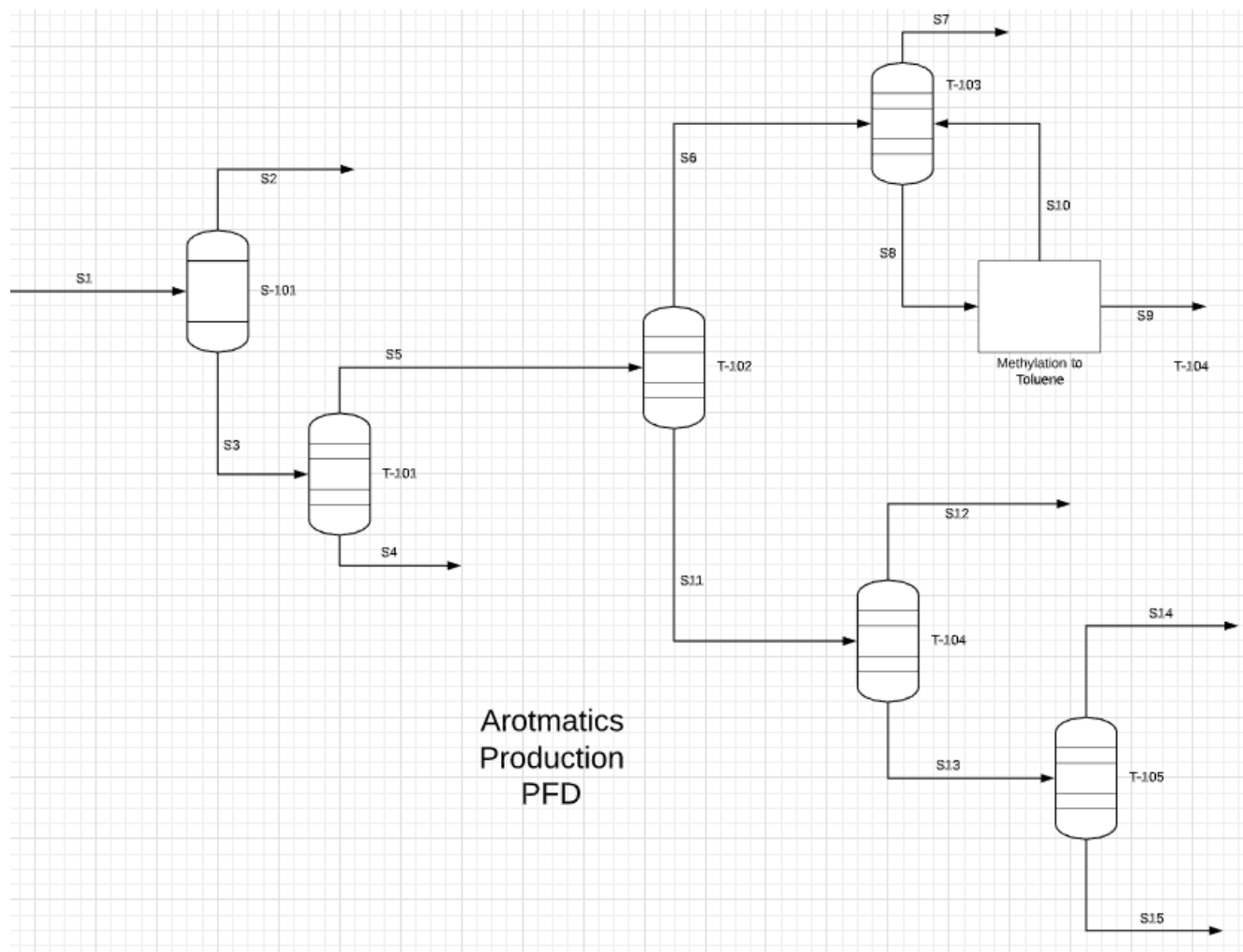


Figure A3-1: PFD of the overall process of the production of para-xylene through the methylation of toluene.

Equipment Descriptions

S-101: Stabilizer Unit – A unit that separates fuel gas from the reformat.

T-101: C₉⁺ Splitter – A adsorption column that separates out C₉⁺ aromatics from the mixture of BTX and ethyl benzene.

T-102: Fractionation Column – A distillation column that separates benzene and toluene from a mixture of ethyl benzene and xylenes.

T-104: C₉⁺ and Ethyl Benzene Splitter – A distillation column that separates out the remaining C₉⁺ aromatics and ethyl benzene from the mixture of xylenes.

T-105: Paraxylene Separator – A separation unit that separates out para-xylene, meta-xylene, and ortho-xylene.

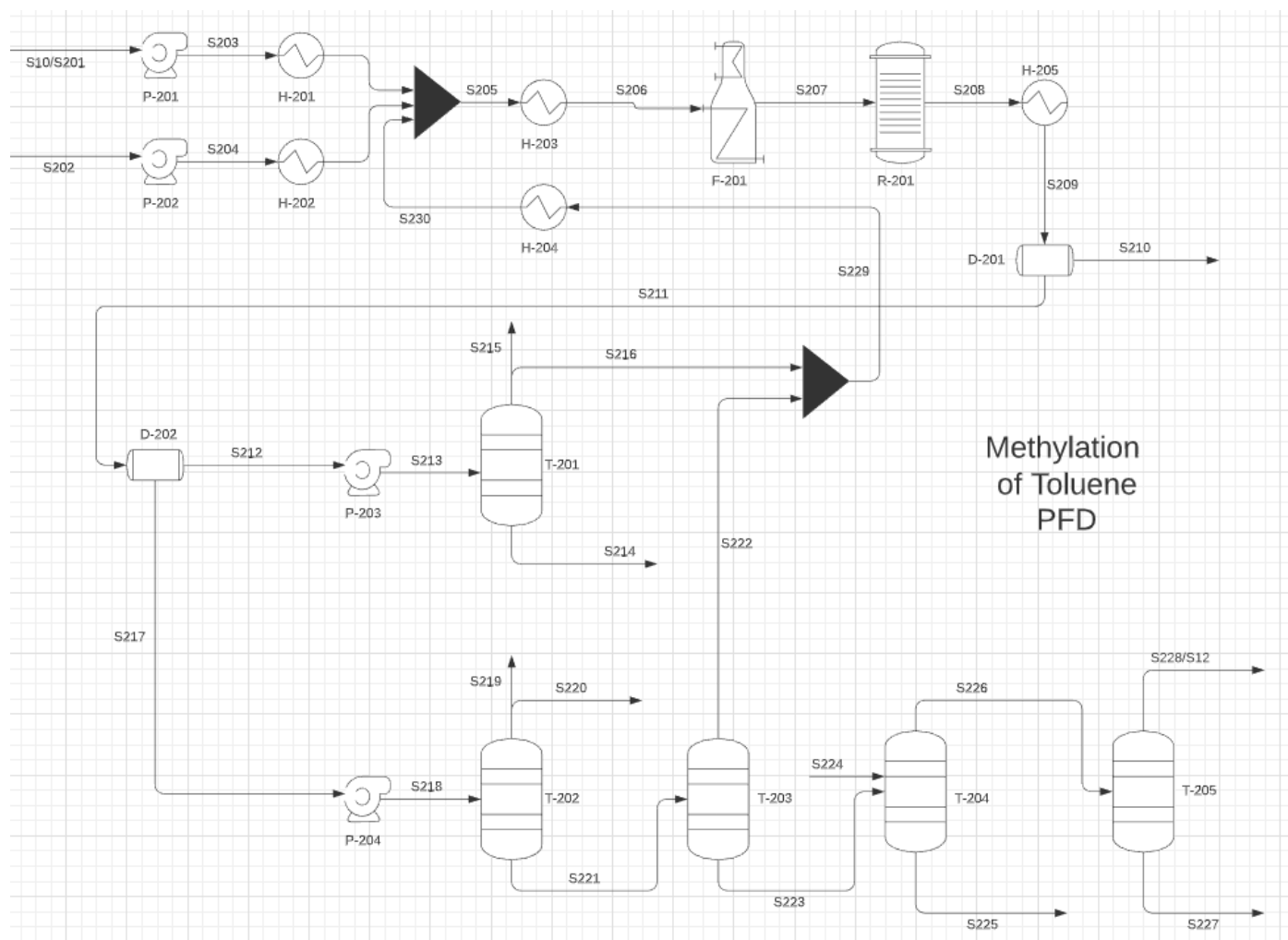


Figure A3-2: PFD of the methylation of toluene unit.

Table A3-1: Equipment list and description of the equipment used in the methylation of toluene process.

Equipment Name	Description	Equipment Name	Description	Equipment Name	Description
P-201, 202, 203, 204	Fluid Pumps	T-201, 202, 203, 204, 205	Distillation Columns	R-201	Packed Bed Reactor
H-201, 202, 203, 204, 205, F-201	Heat Exchanger	D-201	Flash Vessel	D-202	Decanter Vessel

F-201 Furnace: The furnace preheats the feed to the reactor (S206).

R-201 Packed Bed Reactor: Packed bed reactor F-201 is the main block of the process flow sheet. Feed to the reactor is a mixture of fresh methanol, toluene and a methanol recycle stream S206. The combined feed stream (S206) flows into the reactor. The reactor effluent, which is mainly para-xylene, flows out in stream S207.

Cooler H-205: Heat exchanger H-205 cools down the reactor effluent stream S207. The effluent stream S207 is in the gas phase and contains a mixture unreacted toluene and methanol, para-xylene, and reaction by-products (i.e. water, benzene, and gaseous hydrocarbons).

Flash Vessel D-201: Flash vessel D-201 flashes the cooled two-phase stream S209 at a pressure of 2 bar to separate most of the ethylene in the overhead stream S210. 85% of the ethylene fed is separated in the overhead stream S210. The liquid stream (S209) from D-201 goes to decanter D-202.

Decanter D-202: This is a decanter that separates the two liquid phases of S211. Stream S211 contains a mixture of water and immiscible non-polar organic compounds such as toluene and xylene. Methanol is soluble in the water and organic phases. Since toluene and water form an azeotrope, the two compounds are separated before entering the distillation stage.

Methanol Column T-201: The rich water stream from Decanter D-202 contains a large amount of methanol. This methanol needs to be separated and recycled back to the reactor. The distillation column T-201 is used for this purpose. Methanol as a light key exits at the top in the distillate stream

S215. S215 is sent to the circulation heater H-204. Water is obtained in the bottom stream S214 and sent out of the process for further processing or waste.

Purge Column T-202: The organic rich stream S217 in decanter D-202 mainly contains toluene and xylene. Before separating toluene and xylene, we need to separate benzene and other light components to make our recycle stream rich in toluene. This will also have a purifying effect on these light components so that they will not accumulate in a continuous process.

Toluene Column T-203: The bottom stream S221 of T-202 contains toluene and xylene. T-203 separates toluene from xylene. Toluene was obtained as a distillate as a light key. The distillate stream S222 is circulated back to the reactor through a circulation heater H-204. The bottom stream S223 from T-203 contains 97.5% para-xylene, the rest being o-xylene, m-xylene and toluene.

Reactive Distillation Column T-204: The reactive distillation column separates para-xylene from the other xylene isomers.

Para-Xylene Column T-205: The para-xylene column separates out a para-xylene product with 99.5% purity and the distillate is a mixture of left-over BTX, methanol, water, and C9+ aromatics.

Recycle Heater H-204: This is a circulating flow heater. The two streams S216 and S222 are methanol and toluene recycle streams, respectively. They were mixed and passed through the heat exchanger to raise the temperature of the mixed stream S229 to 400 degrees Celsius. The heated stream S230 is at a temperature of 400 degrees Celsius and a pressure of 3 bar. The heat load of the heater is 18872.6 kW.

A4.) – PFD Material Balance

Table A4.1 – Material balance of the overall process of producing para-xylene from reformat with the methylation of toluene process.

	S1		S2		S3		S5		S4		S6		S11	
Species	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)
Propane	1.0%	15272	100.0%	15272	0.0%	0	0.0%	0	0.0%	0	0.0%	0	0.0%	0
Benzene	10.1%	154601	0.0%	0	10.2%	154601	13.4%	153055	0.4%	1546	26.4%	151525	0.3%	1531
Toluene	27.8%	424493	0.0%	0	28.1%	424493	36.7%	420248	1.2%	4245	72.6%	416046	0.7%	4202
ETHYLBENZENE	6.6%	100688	0.0%	0	6.7%	100688	8.7%	99681	0.3%	1007	0.2%	997	17.3%	98684
p-XYLENE	6.8%	104167	0.0%	0	6.9%	104167	9.0%	103126	0.3%	1042	0.2%	1031	17.9%	102094
m-XYLENE	15.2%	232427	0.0%	0	15.4%	232427	20.1%	230102	0.6%	2324	0.4%	2301	39.9%	227801
o-XYLENE	8.9%	135291	0.0%	0	8.9%	135291	11.7%	133938	0.4%	1353	0.2%	1339	23.2%	132599
C9+ (TMB)	23.6%	360299	0.0%	0	23.8%	360299	0.3%	3603	96.9%	356696	0.0%	36	0.6%	3567
SUM:	100.0%	1527238	0.0%	15272	100.0%	1511966	100.0%	1143754	100.0%	368212	100.0%	573275	100.0%	570479

	S7		S8		S8' (Addition of Methanol)		S10		S9		S12		S13	
Species	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)
Water	0.0%	0	0.0%	0	0.0%	0	78.2%	78153	0.2%	789	0.0%	0	0.0%	0
Methanol	0.0%	0	0.0%	0	25.5%	143226	2.8%	2836	0.0%	29	0.0%	0	0.0%	0
Benzene	97.3%	150009	0.4%	1515	0.3%	1515	1.5%	1500	0.0%	15	1.4%	1515	0.0%	15
Toluene	2.7%	4160	98.3%	411885	73.3%	411885	8.2%	8155	0.0%	82	3.7%	4160	0.0%	42
ETHYLBENZENE	0.0%	10	0.2%	987	0.2%	987	1.0%	977	0.0%	10	87.6%	97697	0.2%	987
p-XYLENE	0.0%	10	0.2%	1021	0.2%	1021	4.7%	4661	99.8%	461427	0.9%	1021	22.0%	101073
m-XYLENE	0.0%	23	0.5%	2278	0.4%	2278	2.3%	2255	0.0%	23	2.0%	2278	49.1%	225523
o-XYLENE	0.0%	13	0.3%	1326	0.2%	1326	1.3%	1313	0.0%	13	1.2%	1326	28.6%	131273
C9+ (TMB)	0.0%	0	0.0%	36	0.0%	36	0.0%	35	0.0%	0	3.2%	3531	0.0%	36
SUM:	100.0%	154227	100.0%	419048	100.0%	562274	100.0%	99885	100.0%	462388	100.0%	111529	100.0%	458949

	S14		S15	
Species	wt%	m (kg/hr)	wt%	m (kg/hr)
Water	0.0%	0	0.0%	0
Methanol	0.0%	0	0.0%	0
Benzene	0.0%	0	0.0%	15
Toluene	0.0%	0	0.0%	42
ETHYLBENZENE	0.0%	10	0.3%	977
p-XYLENE	96.5%	100063	0.3%	1011
m-XYLENE	2.2%	2255	62.8%	223268
o-XYLENE	1.3%	1313	36.6%	129960
C9+ (TMB)	0.0%	0	0.0%	35
SUM:	100.0%	103641	100.0%	355308

Table A4.2 – Material balance of the methylation of toluene to paraxylene process.

	S201		S202		S206		S208		S212		S216		S214	
Species	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)
Water	0.0%	0	0.0%	0	0.0%	0	14.0%	78942	91.1%	78153	9.4%	782	99.9%	77371
Methanol	0.0%	0	100.0%	143226	25.5%	143226	0.5%	2865	3.3%	2836	33.6%	2808	0.0%	28
Benzene	0.4%	1515	0.0%	0	0.3%	1515	0.3%	1515	0.0%	15	0.2%	15	0.0%	0
Toluene	98.3%	411885	0.0%	0	73.3%	411885	1.5%	8238	0.1%	82	1.0%	82	0.0%	1
ETHYLBENZENE	0.2%	987	0.0%	0	0.2%	987	0.2%	987	0.0%	10	0.1%	10	0.0%	0
p-XYLENE	0.2%	1021	0.0%	0	0.2%	1021	82.9%	466088	5.4%	4661	55.3%	4614	0.1%	47
m-XYLENE	0.5%	2278	0.0%	0	0.4%	2278	0.4%	2278	0.0%	23	0.3%	23	0.0%	0
o-XYLENE	0.3%	1326	0.0%	0	0.2%	1326	0.2%	1326	0.0%	13	0.2%	13	0.0%	0
C9+ (TMB)	0.0%	36	0.0%	0	0.0%	36	0.0%	36	0.0%	0	0.0%	0	0.0%	0
SUM:	100.0%	419048	0.0%	143226	100.0%	562274	100.0%	562274	100.0%	85793	100.0%	8346	100.0%	77448

	S217		S225		S226		S227		S228	
Species	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)	wt%	m (kg/hr)
Water	0.2%	789	0.1%	8	0.2%	782	5.2%	774	0.0%	8
Methanol	0.0%	29	0.0%	0	0.0%	28	0.2%	28	0.0%	0
Benzene	0.3%	1500	0.2%	15	0.3%	1485	9.9%	1470	0.0%	15
Toluene	1.7%	8155	0.9%	82	1.7%	8074	53.7%	7993	0.0%	81
ETHYLBENZENE	0.2%	977	10.5%	967	0.0%	10	0.1%	10	0.0%	0
p-XYLENE	96.8%	461427	49.9%	4614	97.8%	456812	30.7%	4568	100.0%	452244
m-XYLENE	0.5%	2255	24.1%	2233	0.0%	23	0.2%	22	0.0%	0
o-XYLENE	0.3%	1313	14.0%	1300	0.0%	13	0.1%	13	0.0%	0
C9+ (TMB)	0.0%	35	0.4%	35	0.0%	0	0.0%	0	0.0%	0
SUM:	100.0%	476480	99.9%	9253	100.0%	467227	100.0%	14879	100.0%	452348

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